

New multipartite entanglement measure and its holographic dual

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(Received 30 August 2022; accepted 24 October 2022; published 2 December 2022)

In this paper we define a natural generalization of the von Neumann entropy to multiple parties that is symmetric with respect to all the parties. We call this measure multi-entropy. We show that for conformal field theories with holographic duals, the multi-entropy is computed by the area of an appropriate “soap-film” anchored on the boundary. We conjecture the quantum version of this prescription that takes into account the subleading corrections in G_N .

DOI: [10.1103/PhysRevD.106.126001](https://doi.org/10.1103/PhysRevD.106.126001)

I. INTRODUCTION

In recent years, quantum information theoretic notions, such as entanglement entropy [1–6], have been immensely useful in shedding light on some of the important questions pertaining to quantum gravity. In particular, it has helped understand holographic encoding of the gravitational Hilbert space and, in turn, the black-hole entropy [7,8] and the information paradox. However, with the exception of reflected entropy [9] and negativity [10,11], almost all the discussion has been centered on the entanglement between two parties, i.e., on bipartite entanglement. It is expected that knowing multipartite entanglement structure in quantum gravity would refine our understanding of the holographic encoding of gravitational Hilbert space in conformal field theory Hilbert space. To this end, we introduce a multipartite entanglement measure and its holographic dual.

II. THE MEASURE

Let $|\Psi\rangle \in \bigotimes_{a=1}^q \mathcal{H}_a$ be a quantum state of a q -party system. Let $\dim \mathcal{H}_a = d_a$ and $|\alpha_a\rangle, \alpha_a \in \{1, \dots, d_a\}$ be its orthonormal basis. In $|\alpha_a\rangle$ basis, the state $|\Psi\rangle$ is given as

$$|\Psi\rangle = \sum_{\alpha_1=1}^{d_1} \dots \sum_{\alpha_q=1}^{d_q} \psi_{\alpha_1 \dots \alpha_q} |\alpha_1\rangle \otimes \dots \otimes |\alpha_q\rangle. \quad (1)$$

A q -party entanglement measure is the information contained in the wave function ψ (and its complex conjugate) that is invariant under “local unitary transformations.” They are products of unitary operations $\mathcal{U}_a \in U(d_a)$ performed

in individual system a . Naturally, such a measure is obtained by contracting α_a indices for all a . Now α_a is a fundamental index on ψ and an antifundamental index of $\bar{\psi}$. Hence, in order to construct invariants, we need an equal number of ψ ’s and $\bar{\psi}$ ’s. We will call this number the replica number n . We index the replicas by the superscript (i) . In particular, the Hilbert space $\mathcal{H}_a$ of the i th replica is denoted as $\mathcal{H}_a^{(i)}$ and the associated basis as $|\alpha_a^{(i)}\rangle$. The wave function of the i th replica is then $\psi_{\alpha_1^{(i)} \dots \alpha_q^{(i)}}$ and similarly its conjugate is $\bar{\psi}^{\alpha_1^{(i)} \dots \alpha_q^{(i)}}$.

A. Bipartite

Let us consider the following bipartite entanglement measure with a fixed replica number n :

$$\mathcal{Z}_n \equiv \text{Tr}(\rho^n), \quad \text{where } \rho_{\alpha_1^{(1)} \dots \alpha_q^{(1)}}^{\alpha_1^{(2)} \dots \alpha_q^{(2)}} \equiv \psi_{\alpha_1^{(1)} \alpha_2^{(1)}} \bar{\psi}^{\alpha_1^{(2)} \alpha_2^{(2)}}. \quad (2)$$

Here ρ is the density matrix for party 1. With this notation, the squared norm of $|\Psi\rangle$ is $\mathcal{Z}_1$. This measure is related to the familiar n th Renyi entropy S_n as

$$S_n = \frac{1}{1-n} \text{Log}(\mathcal{Z}_n / \mathcal{Z}_1^n). \quad (3)$$

Let us reformulate this measure using permutations acting on replicas. This will be useful while dealing with entanglement in quantum field theories, using twist operators,

$$\mathcal{Z}_n = (\psi_{\alpha_1^{(1)} \alpha_2^{(1)}} \psi_{\alpha_1^{(2)} \alpha_2^{(2)}} \dots) (\bar{\psi}^{\alpha_1^{(1)} \alpha_2^{(\sigma_1)}} \bar{\psi}^{\alpha_1^{(2)} \alpha_2^{(\sigma_2)}} \dots), \quad (4)$$

where σ is a cyclic permutation element acting in the space of replicas $i = 1, \dots, n$ of the second party, i.e., $\sigma(\alpha_2^{(i)}) = \alpha_2^{(i+1)}$ with $\alpha_2^{(n+1)} \equiv \alpha_2^{(1)}$. Because this permutation acts only on the replica copies of party 2, we denote it as σ_2 . *A priori*, it would seem that we should specify two

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permutation elements (σ_1, σ_2) , the first one acting on replicas of party 1 and the other acting on replicas of party 2 to get the most general index contraction, i.e.,

$$(\psi_{\alpha_1^{(1)}\alpha_2^{(1)}}\psi_{\alpha_1^{(2)}\alpha_2^{(2)}}\dots)(\bar{\psi}_{\alpha_1^{(\sigma_1^{-1})}\alpha_2^{(\sigma_2^{-1})}}\bar{\psi}_{\alpha_1^{(\sigma_1^{-2})}\alpha_2^{(\sigma_2^{-2})}}\dots). \quad (5)$$

However, we can always relabel replicas of $\bar{\psi}$ so that one of the σ 's, say σ_1 , is brought to id form. In other words, (σ_1, σ_2) and $(\sigma_1 g, \sigma_2 g)$, $g \in \mathbb{S}_n$ yield the same measure. We express this as an equivalence relation,

$$(\sigma_1, \sigma_2) \sim (\sigma_1, \sigma_2)g. \quad (6)$$

There is yet another equivalence, namely under simultaneous relabeling of replica copies of *both* ψ 's and $\bar{\psi}$'s. Such a relabeling leads to conjugation of both σ_a by some $h \in \mathbb{S}_n$. We express this as the equivalence relation,

$$(\sigma_1, \sigma_2) \sim h^{-1}(\sigma_1, \sigma_2)h. \quad (7)$$

If we use Eq. (6) to set $\sigma_1 = \text{id}$, then thanks to Eq. (7), the bipartite entanglement measure depends only on the conjugacy class of σ_2 . Conjugacy class in $\mathbb{S}_n$ is specified by cycle decomposition $\{p_k\}$, where p_k is the number of k -cycles such that $\sum_k k p_k = n$. For this class, it is easy to see that the resulting measure (5) is

$$\prod_k (\text{Tr}(\rho^k))^{p_k} = \prod_k (\mathcal{Z}_k)^{p_k}. \quad (8)$$

This shows that the ring of bipartite measures is generated by $\mathcal{Z}_k$'s.

B. Multipartite

Following this discussion, we can write a general multipartite measure by giving permutation elements $(\sigma_1, \dots, \sigma_q)$ with equivalence relations

$$(\sigma_1, \dots, \sigma_q) \sim (\sigma_1, \dots, \sigma_q) \cdot g, \quad (9)$$

$$(\sigma_1, \dots, \sigma_q) \sim h^{-1}(\sigma_1, \dots, \sigma_q)h. \quad (10)$$

Unlike the bipartite case, where Renyi entropies are the only independent measures of entanglement, for the case of three or higher number of parties, the number of measures increase exponentially or faster with the replica number n . In this paper, we will not concern ourselves with general q -party measures but rather focus on a particular family of measures labeled by an integer that is symmetric under the exchange of all the parties.

As in the bipartite case, Eq. (9) can be used to set $\sigma_1 = \text{id}$. But this ‘‘gauge fixing’’ obscures the symmetry between all the parties as it treats party 1 differently from others. So we will not use the gauge freedom (9) and (10) just yet and specify all σ_a 's. We will do so with n^q replicas.

We index the replicas with a q -dimensional index vector $(i_1, \dots, i_q)$, where each $i_a = \{1, \dots, n\}$. The permutation element σ_a is defined as the cyclic element acting only on i_a :

$$\sigma_a \cdot (\dots, i_a, \dots) = (\dots, i_a + 1, \dots), \quad a = 1, \dots, q. \quad (11)$$

As an element of the permutation group $\mathbb{S}_{n^q}$, its conjugacy class is $p_n = n^{q-1}$ with all other $p_k = 0$. It is also clear that this measure is symmetric in all the q parties.

Now we use the gauge freedom (9) with $g = \sigma_1^{-1}$. This gives us an equivalent set of permutation elements,

$$(\text{id}, \sigma_2 \alpha_1^{-1}, \dots, \sigma_q \alpha_1^{-1}). \quad (12)$$

Note that the action of any of these permutations on $(i_1, \dots, i_q)$ keeps the sum $\sum_a i_a$ invariant. As a result, their action on n^q replicas splits into n orbits of n^{q-1} elements with $\sum_a i_a = \text{constant} \pmod{n}$. Each orbit gives rise to the same invariant of ψ 's and $\bar{\psi}$'s. It is convenient to work with a single orbit, say with $\sum_a i_a = 0 \pmod{n}$. For this orbit, we can use the relabeling freedom (10) to set the replica index i_1 to 1. With this gauge fixing, we get a convenient presentation of the rest of the permutation elements $(\sigma_2 \alpha_1^{-1}, \dots, \sigma_q \alpha_1^{-1}) \equiv (\hat{\sigma}_2, \dots, \hat{\sigma}_q)$,

$$\hat{\sigma}_a \cdot (\dots, i_a, \dots) = (\dots, i_a + 1, \dots), \quad a = 2, \dots, q. \quad (13)$$

We have denoted the gauge fixed permutation elements as $\hat{\sigma}_a$ to avoid confusion with ungauged fixed permutation elements σ_a . The index set in Eq. (13) starts from i_2 because i_1 has already been set to 1. We denote the measure defined by $\hat{\sigma}_a$ elements in (13) as $\mathcal{Z}_n^{(q)}$ and define Renyi multi-entropy $S_n^{(q)}$ as

$$S_n^{(q)} \equiv \frac{1}{1-n} \text{Log}(\mathcal{Z}_n^{(q)} / (\mathcal{Z}_1^{(q)})^{n^{q-1}}). \quad (14)$$

Here the factor $(\mathcal{Z}_1^{(q)})^{n^{q-1}}$ serves to normalize the state. This is the family of q -partite measures that we are interested in. Let us pause for a moment to consider properties of the set $\{\hat{\sigma}_a : a = 2, \dots, q\}$:

- (i) $\hat{\sigma}_a$'s have the same equivalence class given by the cycle $p_n = n^{q-2}$ with all other $p_k = 0$.
- (ii) $\hat{\sigma}_a^{-1} \hat{\sigma}_b$ for $a \neq b$ also have the same equivalence class.
- (iii) Together, $\hat{\sigma}_a$'s generate the subgroup $\mathbb{Z}_n^{\otimes q-1}$ of the permutation group.

These observations will be important in the future discussions.

In a way, fixing $\hat{\sigma}_1 = \text{id}$ has a very natural interpretation. If we consider a single ψ and a single $\bar{\psi}$ with i_1 index contraction, then we get the density matrix on $\mathcal{H}_2 \otimes \dots \otimes \mathcal{H}_q$. This density matrix has the index structure

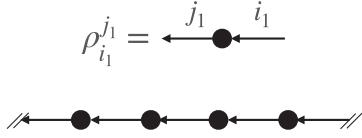


FIG. 1. Graphical notation for the density matrix for $q = 2$ and the index contraction that gives $\mathcal{Z}_4^{(2)}$.

$$\rho_{\alpha_2^{(i_2)} \dots \alpha_q^{(i_q)}}^{\alpha_2^{(\hat{\sigma}_2 \cdot i_2)} \dots \alpha_q^{(\hat{\sigma}_q \cdot i_q)}}. \quad (15)$$

This means the gauge fixed permutations $\hat{\sigma}_a$'s describe index contractions of this density matrix. Let us see this with two examples, for $q = 2$,

$$\mathcal{Z}_n^{(2)} = \sum \prod_{i_2} \rho_{\alpha_2^{(i_2)}}^{\alpha_2^{(\hat{\sigma}_2 \cdot i_2)}} = \sum \rho_{\alpha_1^{(2)}}^{\alpha_2^{(\hat{\sigma}_2 \cdot 1)}} \rho_{\alpha_2^{(2)}}^{\alpha_2^{(\hat{\sigma}_2 \cdot 2)}} \dots \rho_{\alpha_2^{(\hat{\sigma}_2 \cdot n)}}^{\alpha_2^{(\hat{\sigma}_2 \cdot n)}}.$$

Here sum is over all repeated indices. With $\hat{\sigma}_2$ being a cyclic element as specified by Eq. (13), this is nothing but $\mathcal{Z}_n$ defined earlier. So we have showed $S_n^{(2)} = S_n$. Graphically, denoting ρ as in Fig. 1, we get the graphical representation of $S_n^{(2)}$ as a circular lattice of ρ 's of length n as in Fig. 1. Each directed link in this lattice represents index contraction from a fundamental index to an anti-fundamental index. The advantage of this graphical notation is that it admits straightforward extension to higher number of parties. For instance, in the case of 3-parties, we denote the density matrix as in Fig. 2. Then $\mathcal{Z}_n^{(3)}$ is the index contraction given by the toric lattice of ρ 's with both sides being of length n . This is shown in Fig. 2. Similarly, $\mathcal{Z}_n^{(q)}$ is the index contraction given by $(q-1)$ -dimensional toric lattice with all sides being of length n .

Arguably the most important measure of bipartite entanglement, the von Neumann entropy S , is obtained as a limit,

$$S = \lim_{n \rightarrow 1} S_n. \quad (16)$$

The Renyi multi-entropy $S_n^{(q)}$ is defined such that the limit

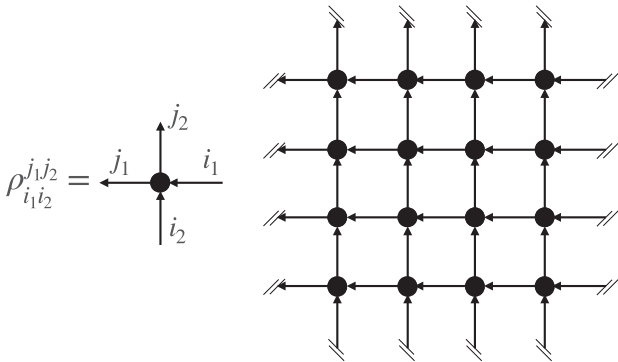


FIG. 2. Graphical notation for the density matrix for $q = 3$ and the index contraction that gives $\mathcal{Z}_4^{(3)}$.

$$S^{(q)} = \lim_{n \rightarrow 1} S_n^{(q)}, \quad (17)$$

called *the* multi-entropy, has a number of nice properties:

- (1) It is symmetric in all the parties.
- (2) For $q = 2$, it reduces to the von Neumann entropy.
- (3) It admits a convenient holographic description.

Properties 1 and 2 are obvious from the discussion so far. In the rest of the paper, we will discuss multi-entropy for holographic theories and describe what we exactly mean by property 3. The multi-entropy can also be computed from $\mathcal{Z}_n^{(q)}$ using the formula

$$S^{(q)} = -\partial_n \log(\mathcal{Z}_n^{(q)} / (\mathcal{Z}_1^{(q)})^{n^{q-1}}) |_{n=1}. \quad (18)$$

We will use this formula and Eq. (17) interchangeably.

C. Example

Although the multi-entropy is amenable to holographic computation, as we will see shortly, it is difficult to compute for finite dimensional quantum systems. This is because multi-entropy is defined only via the replica trick and, in general, it is difficult to compute $\mathcal{Z}_n^{(q)}$ as an analytic function of n . However, for a special class of states that we call generalized Greenberger–Horne–Zeilinger (GHZ) states, the multi-entropy is readily computed. Consider $|\Psi\rangle \in \otimes_{a=1}^q \mathcal{H}_a$ with $\dim \mathcal{H}_a = d$. The generalized GHZ state is defined as

$$|\Psi\rangle_{\text{GHZ}} = \sum_i \lambda_i |i\rangle \otimes \dots \otimes |i\rangle. \quad (19)$$

Here $|i\rangle$ is an orthonormal basis in $\mathcal{H}_a$ for all a . The state is normalized so $\sum_i |\lambda_i|^2 = 1$. It is easy to compute $\mathcal{Z}_n^{(q)}$ because the same index i runs through all the contractions. It contributes $|\lambda_i|^{2n^{q-1}}$. Then the q -Renyi entropy is

$$S_n^{(q)} = \frac{1}{1-n} \log \left(\sum_i |\lambda_i|^{2n^{q-1}} \right). \quad (20)$$

Taking the $n \rightarrow 1$ limit,

$$S^{(q)} = (1-q) \sum_i |\lambda_i|^2 \log |\lambda_i|^2. \quad (21)$$

A bipartite state can always be taken to the generalized GHZ form via Schmidt decomposition. Then $|\lambda_i|^2$ are the eigenvalues of the density matrix. It is clear that $S^{(q)}$ agrees with the von Neumann entropy for $q = 2$. We believe that it is extremely important to develop techniques to compute multi-entropy for general states to understand its quantum information theoretic properties.

D. Distinguishing isospectral density matrices

In this section, we will discuss the effectiveness of Renyi multi-entropy in distinguishing multipartite entanglement in two mixed states. For this purpose, we will present density matrices ρ_{12} and σ_{12} on two qubits such that the spectrum of ρ_{12} , ρ_1 and ρ_2 is identical to that of σ_{12} , σ_1 and σ_2 , respectively. Such density matrices are known as isospectral. This example of the isospectral pair is borrowed from [12]. The isospectral property of these density matrices guarantees that no bipartite measure of entanglement can distinguish between these states. However, as we will show, all the Renyi multientropies for these two states are different,

$$\rho_{12} = \frac{1}{3} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \sigma_{12} = \frac{1}{3} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}. \quad (22)$$

The nonzero eigenvalues of ρ_{12} , ρ_1 , ρ_2 (and also for σ_{12} , σ_1 , σ_2) are $\frac{1}{3}, \frac{2}{3}$. The values of the first few Renyi multi-entropies are

	ρ	σ
$S_2^{(3)}$	$\text{Log}(9)$	$\text{Log}(\frac{81}{17})$
$S_3^{(3)}$	$\frac{1}{2} \text{Log}(\frac{6561}{14})$	$\frac{1}{2} \text{Log}(\frac{729}{19})$
$S_4^{(3)}$	$\frac{1}{3} \text{Log}(\frac{14348907}{139})$	$\frac{1}{3} \text{Log}(\frac{43046721}{65537})$

In fact, it is easy to see that ρ_{12} is obtained by tracing out one party in the pure W-state $(|100\rangle + |010\rangle + |001\rangle)/\sqrt{3}$ and σ_{12} is obtained analogously from the pure generalized GHZ state $(|000\rangle + \sqrt{2}|111\rangle)/\sqrt{3}$. The density matrix σ_{12} is separable, while the density matrix ρ_{12} is not. The fact that the (Renyi) multi-entropy distinguishes the two suggests that it can lead to a new separability criterion. We are currently exploring this possibility.

III. MULTI-ENTROPY FROM HOLOGRAPHY

In this section, we will discuss computation of multi-entropy in a D -dimensional conformal field theory $\mathcal{T}$. Let the state $|\Psi\rangle$ be defined on a time symmetric Cauchy slice $\mathcal{R}$ of a D -manifold $\mathcal{M}$. It is given by a Euclidean path integral on the half-space $\mathcal{M}_\Psi$ such that $\partial\mathcal{M}_\Psi = \mathcal{R}$. The dual bra $\langle\Psi|$ is constructed by Euclidean path integral on the other half $\mathcal{M}_{\bar{\Psi}}$. It is obtained from $\mathcal{M}_\Psi$ by reflecting across $\mathcal{R}$. The squared norm of Ψ is the partition function $\mathcal{Z}_\mathcal{M}$ on $\mathcal{M}$. Let us decompose $\mathcal{R}$ into q number of disjoint regions $\mathcal{R}_a$, such that $\cup_a \mathcal{R}_a = \mathcal{R}$. Let the Hilbert space on region $\mathcal{R}_a$ be $\mathcal{H}_a$. We are interested in computing multi-entropy of the state $|\Psi\rangle$ under the decomposition $\otimes_a \mathcal{H}_a$.

For theories that admit a weakly coupled gravity dual, this problem can be addressed holographically.

The replica trick [1] involves working with the tensor product theory $\mathcal{T}^{\otimes n^{q-1}}$ on $\mathcal{M}$. This theory has the discrete symmetry $\mathbb{S}_{n^{q-1}}$. It admits codimension 2 twist defects labeled by elements of this permutation group. For every pair of regions $(\mathcal{R}_a, \mathcal{R}_b)$ that share a boundary, we insert the twist operator $\mathcal{O}_{\hat{\sigma}_a^{-1}\hat{\sigma}_b}$ on the common boundary. The measure $\mathcal{Z}_n^{(q)}$ is then given by the correlation function of these twist operators. Let us denote the resulting replicated manifold as $\mathcal{M}_n$. The correlation function of twist operators is the partition function $\mathcal{Z}_{\mathcal{M}_n}$ on $\mathcal{M}_n$. Using (14) we have

$$S_n^{(q)} = \frac{1}{1-n} \text{Log}(\mathcal{Z}_{\mathcal{M}_n}/(\mathcal{Z}_\mathcal{M})^{n^{q-1}}). \quad (23)$$

Following [13], we will proceed to analyze this problem holographically. Let $\mathcal{B}_n$ be dominant the gravity solutions such that $\partial\mathcal{B}_n = \mathcal{M}_n$. The manifold $\mathcal{B}_n$ is a smooth manifold, e.g., in the case of Einstein gravity with negative cosmological constant, it is of constant negative curvature. The holographic dictionary gives

$$\text{Log}\mathcal{Z}_{\mathcal{M}_n} = -\mathcal{S}_{\text{grav}}(\mathcal{B}_n). \quad (24)$$

Here $\mathcal{S}_{\text{grav}}(\mathcal{X})$ is the gravitational action evaluated on the solution $\mathcal{X}$. For $n = 1$, this gives $\text{Log}\mathcal{Z}_\mathcal{M} = -\mathcal{S}_{\text{grav}}(\mathcal{B})$, $\mathcal{B} \equiv \mathcal{B}_1$.

The background fields on the manifold $\mathcal{M}_n$ enjoy a replica symmetry. This is the symmetry generated by the permutation elements $\hat{\sigma}_a^{-1}\hat{\sigma}_b$ associated to all the twist operators. As remarked earlier, this group is $\mathbb{Z}_n^{\otimes q-1}$. Following [13], we will assume that the dominant bulk solution $\mathcal{B}_n$ enjoys this symmetry. The solution $\mathcal{B}_n$ consists of codimension 2 loci that are invariant under certain subgroups of the replica symmetry group. Some of these loci, called “external,” are anchored at the fixed points on the boundary (these are locations of twist operator insertions on $\mathcal{M}$), while the rest are “internal.” Let us denote the loci that are anchored at the fixed points corresponding to the twist operator $\mathcal{O}_g$ as $\mathcal{L}_g$. They are invariant under the $\mathbb{Z}_n$ subgroup generated by g . Generically, two $\mathcal{L}$ s can merge to form a different $\mathcal{L}$. Merging obeys the algebra $\mathcal{L}_{g_1} \cdot \mathcal{L}_{g_2} \rightarrow \mathcal{L}_{g_1 g_2}$. The internal loci come about because of such merging. Below we will assume that:

(1) Every fixed point locus is of the form $\mathcal{L}_{\hat{\sigma}_a^{-1}\hat{\sigma}_b}$. We will justify this assumption shortly. Thanks to the special property of our permutation elements $\hat{\sigma}_a$ stated below Eq. (14), this means that any locus is invariant under some $\mathbb{Z}_n$ subgroup of the replica symmetry. Every locus $\mathcal{L}_g$ appears in groups of n^{q-2} . This is because, the orbit of the action of replica group $\mathbb{Z}_n^{\otimes q-1}$ consists of n^{q-2} elements as its stabilizer is $\mathbb{Z}_n$. For $q = 2$, the number of elements in the orbit is 1.

We now make use of the replica symmetry in the bulk to construct the orbifold $\tilde{\mathcal{B}}_n \equiv \mathcal{B}_n/(\mathbb{Z}_n)^{q-1}$. Due to symmetry, the classical gravitational actions on the two spaces are related as

$$\mathcal{S}_{\text{grav}}(\mathcal{B}_n) = n^{q-1} \mathcal{S}_{\text{grav}}(\tilde{\mathcal{B}}_n). \quad (25)$$

The orbifold $\tilde{\mathcal{B}}_n$ has a nice property that $\partial\tilde{\mathcal{B}}_n = \mathcal{M}$. A group of n^{q-2} number of $\mathcal{L}_g$ becomes a single conical singularity of opening angle $2\pi/n$ in the orbifold $\tilde{\mathcal{B}}_n$. Let us denote this singularity as $\tilde{\mathcal{L}}_g$. Let us denote the web created by these singularities as $\mathcal{W}$. Consider a codimension 1 slice $\mathcal{C} \in \tilde{\mathcal{B}}_n$ that contains $\mathcal{W}$ and $\partial\mathcal{C} = \mathcal{R}$. There are multiple such slices, and the precise choice does not matter for the following discussion. Every singularity becomes a codimension 1 wall in $\mathcal{C}$, and the web $\mathcal{W}$ yields its chamber decomposition. As we move from $\mathcal{R}_a$ to $\mathcal{R}_b$ through $\mathcal{C}$, we must encounter at least one wall because the permutation elements $\hat{\sigma}_a$ and $\hat{\sigma}_b$ are different. At this stage, we make an assumption about $\mathcal{W}$ that:

- (2) There is no chamber which lies completely in the interior of $\mathcal{C}$.

As a result, we get a one-to-one map between the chambers and boundary regions. Let us denote the chamber adjacent to $\mathcal{R}_a$ as $\mathcal{C}_a$. It has the property $\partial\mathcal{C}_a \cap \mathcal{M} = \mathcal{R}_a$. The web $\mathcal{W}$ consists of only those walls that separate $\mathcal{C}_a$ and $\mathcal{C}_b$ for some (a, b) . Such a wall must be of the type $\tilde{\mathcal{L}}_{\hat{\sigma}_a^{-1}\hat{\sigma}_b}$. The parent $\mathcal{L} \in \mathcal{B}_n$ must also be of the same type. This justifies our assumption 1.

To compute the q -Renyi entropy using Eqs. (24) and (25), we need to evaluate the gravitational action on the orbifold solution $\tilde{\mathcal{B}}_n$. In what follows, we will specialize to the case of Einstein gravity. The orbifold solution is a smooth solution of Einstein's equations with constant negative curvature, except at $\tilde{\mathcal{L}}$, where it has a conical singularity of opening angle $2\pi/n$. The gravitational action for this solution is computed as follows [13]. We excise a small tubular neighborhood, say of radius a , around the conical singularity $\tilde{\mathcal{L}}$. The gravitational action comes from the Gibbons-Hawking-York (GHY) term evaluated on the resulting boundary. This contribution is extensive in the codimension 2 area of $\tilde{\mathcal{L}}$. In the limit $a \rightarrow 0$, the action takes the form [14], $\frac{1-n}{4nG_N} \int dy^{D-1} \sqrt{h}$. Here y is a coordinate along $\tilde{\mathcal{L}}$ and h is the induced metric on $\tilde{\mathcal{L}}$. To compute the equations that are obeyed by the singular locus, it is useful to introduce a cosmic brane with the same action supported on the singularity,

$$\mathcal{S}_{\text{br}}^{(n)} = \frac{1-n}{4nG_N} \int dy^{D-1} \sqrt{h} = \frac{1-n}{4nG_N} A. \quad (26)$$

The y integral is simply the area A of the brane web $\mathcal{W}$. The solution is then computed by solving the equations coming from the action $\mathcal{S}_{\text{grav}} + \mathcal{S}_{\text{br}}^{(n)}$. It is known [15,16] that such

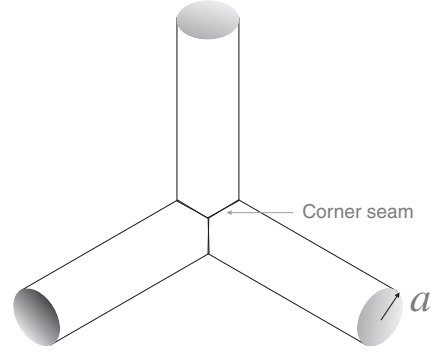


FIG. 3. Top view of the junction of tubular neighborhoods of three $\tilde{\mathcal{L}}$'s.

a cosmic brane action indeed gives rise to a conical singularity of opening angle $2\pi/n$ in the two transverse directions as desired. Once the solution is found by extremizing $\mathcal{S}_{\text{grav}} + \mathcal{S}_{\text{br}}^{(n)}$, we need to evaluate only the action $\mathcal{S}_{\text{grav}}$ on the solution. This is because the cosmic brane is not actually present at the singularity but merely used as a trick to model the singularity. The novelty in the multipartite case is that multiple $\tilde{\mathcal{L}}_g$'s can meet each other at higher codimensional loci. To accommodate such a meeting, we simply let the corresponding cosmic branes meet.

A priori, there can be additional terms in the brane action supported only at the meeting locus. To compute such terms, we again excise the tubular neighborhood of $\tilde{\mathcal{L}}$'s and focus near their higher codimensional junction. As multiple $\tilde{\mathcal{L}}$'s meet, their corresponding tubular neighborhoods also meet forming corners as shown in Fig. 3. For the gravitational variational principle to be well defined, we need to add the so-called Hayward term at the corner just as the way we add the GHY term on the boundary,

$$\mathcal{S}_{\text{Hayward}} = -\frac{1}{8\pi G_N} \int d\xi^{D-2} (\theta - \pi) \sqrt{\gamma}. \quad (27)$$

Here θ is the angle subtended by the two boundaries at the corner, ξ is the coordinate along the corner, and γ is the induced metric on the corner. The contribution of the Hayward term is extensive along the codimension 3 meeting locus, and the proportionality constant is computed by integrating the Hayward term along the one-dimensional corner seam of the tubular neighborhood as shown in Fig. 3. This integral is proportional to a and vanishes as we take $a \rightarrow 0$. This had to be the case on dimensional grounds because the contribution at the meeting locus must take the form $\sim \frac{1}{G_N} \ell_{\text{meeting}}^{(3)} \ell_{\text{scale}}$. Here $\ell_{\text{meeting}}^{(3)}$ is the length (codimension 3) of the meeting locus in $\mathcal{W}$, and ℓ_{scale} is some length scale needed to obtain a dimensionless answer. The only length scale that could serve this purpose is a [17], which we take to 0. This

argument also shows that there is no extra contribution to the action even at higher codimension meeting loci.

Finding the solution to the theory $\mathcal{S}_{\text{grav}} + \mathcal{S}_{\text{br}}^{(n)}$ for general n is still a daunting task (see Refs. [14,18,19] for computation of bipartite Renyi entropy). But to compute multi-entropy, only the limit $n \rightarrow 1$ is relevant. As the only n dependence appears in the coefficient in $\mathcal{S}_{\text{br}}^{(n)}$, the solution can be analytically continued away from the n integer. Moreover, in the limit $n \rightarrow 1$, the tension of the brane goes to zero, and the solution can be found in the probe limit as brane web configuration that extremizes $\mathcal{S}_{\text{br}}$ in the fixed background $\mathcal{B}$. The solution obeys the equation of motion $\delta_{g_{\mu\nu}}(\mathcal{S}_{\text{grav}} + \mathcal{S}_{\text{br}}^{(n)}) = 0$. In particular,

$$\partial_n \mathcal{S}_{\text{grav}} = -\partial_n \mathcal{S}_{\text{br}}^{(n)} = \frac{A}{4G_N}. \quad (28)$$

This, along with Eqs. (17), (23), and (24), shows that the multi-entropy is given a simple formula

$$S^{(q)} = \frac{A(\mathcal{W})}{4G_N}. \quad (29)$$

Here $A(\mathcal{W})$ is the area of the minimal brane web $\mathcal{W}$ in $\mathcal{B}$. The brane web $\mathcal{W}$ obeys the topological conditions:

- (1) $\mathcal{W}$ is anchored at the boundaries of all the regions $\mathcal{R}_a$'s.
- (2) $\mathcal{W}$ contains subwebs that are homologous to all the regions $\mathcal{R}_a$'s.

The second condition is the reformulation of the statement that between any two chambers $\mathcal{C}_a$ and $\mathcal{C}_b$ there must be at least one wall. As the solution minimizes the area subject to these conditions, it does not allow any chamber that lies completely in the interior of $\mathcal{C}$. This justifies our assumption 2.

The brane web in question is extremely familiar in $D = 4$. In the three-dimensional Cauchy slice $\mathcal{C}$ it resembles a soap film in hyperbolic space anchored on a given “wire frame” at infinity. Specifying the anchor does not specify the soap film uniquely, but when combined with the homology condition and the global minimum condition, it does. The close analogy to soap films leads us to call our prescription, the soap-film prescription. To describe the soap film, it is convenient to introduce the label $\tilde{\mathcal{L}}^{(k)}$ for the special locus of codimension k . It is defined inductively as the meeting locus of $\tilde{\mathcal{L}}^{(k-1)}$ with $\tilde{\mathcal{L}}^{(2)} \equiv \tilde{\mathcal{L}}$. For $D = 4$ it is known that:

- (i) Three $\tilde{\mathcal{L}}^{(2)}$'s meet at $\tilde{\mathcal{L}}^{(3)}$ at an angle $2\pi/3 = \cos^{-1}(-1/2)$.
- (ii) Four $\tilde{\mathcal{L}}^{(3)}$'s meet at $\tilde{\mathcal{L}}^{(4)}$ at an angle $\cos^{-1}(-1/3)$.

These are known as Plateau's laws of soap film. In dimension D we expect $k+1$ of $\tilde{\mathcal{L}}^{(k)}$'s to meet at $\tilde{\mathcal{L}}^{(k+1)}$ at an angle $\cos^{-1}(-1/k)$ and so on until we get to $\tilde{\mathcal{L}}^{(D)}$.

We have verified our proposal for $2D$ conformal field theories with large central charge. These calculations will appear in an accompanying paper.

IV. GENERALIZATIONS

In this section we will comment on the generalizations of our soap-film prescription in various directions.

A. Covariant prescription

The covariant generalization of the Ryu-Takayanagi formula was presented in [3]. It was argued there that the entanglement entropy is given by area of the extremal surface rather than the minimal one. This prescription was later rephrased as a maximin prescription in [20], where the surface in question is obtained first by minimizing the area on some achronal slice Σ and then by maximizing the area with respect to the variation of Σ . In case there are multiple extremal solutions, we pick the one with the minimum area as that corresponds to the most dominant solution.

The soap-film prescription proposed here admits a natural covariant generalization along the lines of [3] and [20]. We conjecture that the covariant multi-entropy is obtained as the area of the minimal extremal soap film rather than the globally minimal one and that this choice is equivalent to the one obtained by the maximin prescription of [20].

B. Higher derivative gravity

For higher derivative gravitational theories, we do expect a nontrivial contribution localized at the higher codimensional meeting loci $\tilde{\mathcal{L}}^{(k)}$'s because the dimensional analysis argument presented in Sec. III ceases to be valid as ℓ_{scale} can be provided by the inverse mass scale of the higher derivative corrections. It would be interesting to compute these terms in general higher derivative theory of gravity.

C. Quantum corrections

Quantum correction to the multi-entropy can be found using the replica trick in the bulk. We consider n^{q-1} replica copies of the orbifold $\tilde{\mathcal{B}}_n$ and insert the bulk twist operators $\mathcal{V}_g$ at $\tilde{\mathcal{L}}_g$. This has the effect of reversing the orbifold and give back the geometry $\mathcal{B}_n$. The partition function on the replicated manifold is then simply the partition function $\mathcal{Z}_n^{(q)}$. This is the partition function that goes into the calculation of the multi-entropy. If the replica trick was performed on the bulk geometry $\mathcal{B}$, it would have given the bulk multi-entropy $S_{\text{bulk}}^{(q)}(\mathcal{W})$ corresponding to the chamber decomposition $\mathcal{C}_a$ directly, however, because the replica trick was performed on $\tilde{\mathcal{B}}_n$, it is not obvious that what we get is $S_{\text{bulk}}^{(q)}(\mathcal{W})$. This situation is similar to the bipartite case [21] (see also Ref. [22]). There, the difference between the two quantities is captured by changing the classical solution by $\mathcal{O}(G_N)$ to account for the one-loop expectation value of the stress tensor. This changes the area and hence the entanglement entropy by $\mathcal{O}(1)$. We expect a similar

formula to give the subleading correction to the multi-entropy

$$S^{(q)} = \frac{\langle \hat{A}(\mathcal{W}) \rangle}{4G_N} + S_{\text{bulk}}^{(q)}(\mathcal{W}) + \text{c.t.} \quad (30)$$

Here $\hat{A}(\mathcal{W})$ is the area operator of the soap film and c.t. are the counterterms that render $S_{\text{bulk}}^{(q)}(\mathcal{W})$ finite.

Following [6] and [23], we conjecture a formula that is valid to all orders in $1/G_N$ perturbation theory: Multi-entropy is given by the above formula, but $\mathcal{W}$ is not the ordinary area extremizing soap film but rather the “quantum extremal soap film,” i.e., the soap film that extremizes the combination

$$\frac{A(\mathcal{W})}{4G_N} + S_{\text{bulk}}^{(q)}(\mathcal{W}). \quad (31)$$

In analogy with the bipartite case, we call this prescription the quantum extremal soap-film prescription.

ACKNOWLEDGMENTS

We would like to thank Akshay Gadde, Gautam Mandal, Shiraz Minwalla, Piyush Shrivastava, Pushkal Shrivastava, and Sandip Trivedi for interesting discussions. We are particularly indebted to Onkar Parrikar for stimulating discussions and insightful comments. We would also like to thank Shraiyan Jain for collaboration on related projects. This work is supported by the Infosys Endowment for the study of the Quantum Structure of Spacetime and by the SERB Ramanujan fellowship. We acknowledge support of the Department of Atomic Energy, Government of India, under Project Identification No. RTI 4002.

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